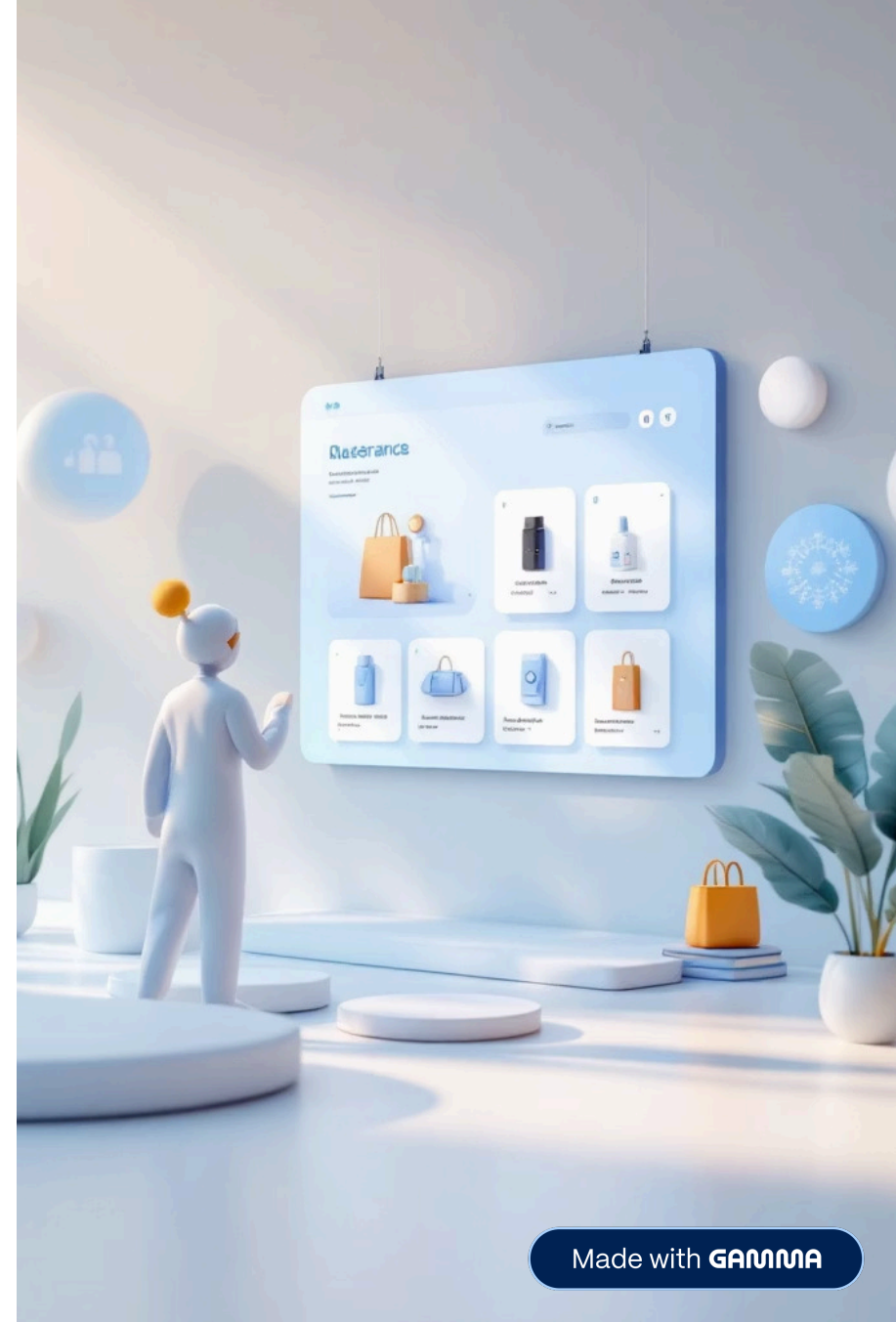
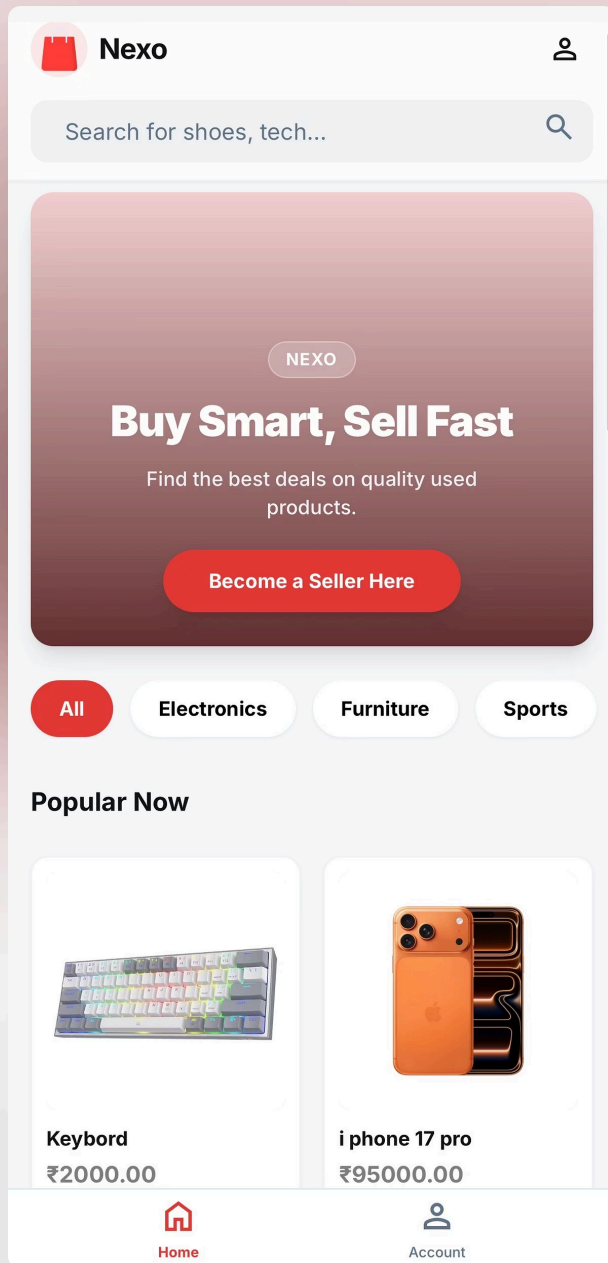


Predictive Analytics Based Student Marketplace

NEXO -AI-Powered Second-Hand Student Marketplace





PROJECT OVERVIEW

NEXO: Your Smart Second-Hand Marketplace

NEXO is an innovative second-hand product marketplace crafted exclusively for college students. It connects students looking to buy or sell used academic materials, gadgets, and accessories.

- Specialized for students: Focuses on the unique needs of the college community.
- Affordable essentials: Facilitates access to cost-effective books and tech.
- Predictive analytics: Enhances search and product visibility for a seamless experience.
- Smarter discovery: Moves beyond basic listings to intelligent product matching.

Robust Architecture for Intelligent Solutions

Frontend

- React.js: Dynamic and responsive user interfaces.
- Tailwind CSS: Utility-first styling for a sleek, modern look.

Backend

- Django: Powerful, high-level Python web framework.
- REST API: Seamless data exchange and integration.

AI/ML Tools

- Pandas & NumPy: Efficient data manipulation and numerical operations.
- Matplotlib & Seaborn: Advanced data visualization.
- NLTK & re: Natural Language Processing for text analysis.
- Joblib: For efficient model persistence.



AI AT THE CORE

Intelligent Models Driving NEXO



Product Engagement Ranking

Our AI model dynamically ranks products based on diverse user interactions such as clicks, views, and wishlist additions. This ensures that the most relevant and popular items gain optimal visibility, enhancing user experience and driving sales.



Search Success Prediction

This model predicts the likelihood of a user's search query yielding satisfactory results. If the success probability is low, it intelligently suggests refined search terms or alternative products, preventing user frustration and improving discovery.

NEXO: Today's Features, Tomorrow's Vision

Key Features

- **AI-powered ranking system:** Ensures relevant products are seen.
- **Smart search optimization:** Guarantees effective product discovery.
- **Student-focused marketplace:** Tailored for university needs.
- **Predictive analytics engine:** Drives intelligent platform functions.
- **Clean and modern UI:** Intuitive and engaging user experience.

Future Scope

- **Price prediction model:** Helps users identify fair pricing.
- **Personalized recommendations:** Custom product suggestions based on user behavior.
- **Real-time demand analysis:** Optimizes inventory and listing strategies.